

TaxaHabitat

Habitat assignment and spatial quality control for the TaxaID ecosystem. Classifies species into habitat categories using LLM-based biological consensus, assigns habitats to sampling sites, and flags spatial outliers.

Habitat is a key predictor of which species are plausible at a sampling location. Incorrect habitat classification leads to false positives when species from the wrong habitat receive inflated priors. Ideally, a knowledgeable user assigns each taxon to its habitat. When the species list is long, LLMs can do a reasonable job of classifying species into habitat categories. As with any LLM output, users should review the results. `flag_habitat_inconsistencies()` provides an interactive map that makes errant classifications easy to spot and correct.

A key feature of TaxaHabitat is the ability to review and proof occupancy data. Even well-curated data like GBIF have a high frequency of location errors. By mapping points by habitat type, users can easily view which observations have incorrect coordinates. The function `review_spatial_flags(occurrences_flagged)` is designed to flag errant points for removal before model building begins. TaxaHabitat thus can be a standalone database QAQC for biodiversity databases.

Habitat Schemes

Scheme	Categories	Use case
3-category (default)	Marine / Freshwater / Terrestrial	Most eDNA studies
IUCN Level 1	18 IUCN habitat categories	Fine-grained habitat mapping
Custom	User-defined	Specialized study designs

Installation

```
# Requires TaxaTools (foundation package)
devtools::install("path/to/TaxaTools")
devtools::install("path/to/TaxaHabitat")
```

Quick Start

```
library(TaxaHabitat)

# 1. Build an LLM prompt for habitat classification
prompt <- build_habitat_prompt(
  c("Fundulus parvipinnis", "Cottus asper", "Anas platyrhynchos")
)

# 2. Submit to an LLM provider
raw_text <- TaxaTools::prompt_api(prompt)

# 3. Parse response into per-species habitat weights
habitat_weights <- parse_hierarchical_habitat_response(raw_text, prompt)
# taxon_name      Marine_weight Freshwater_weight Terrestrial_weight
# Fundulus parvipinnis 0.85          0.15          0.0
# Cottus asper      0.0            1.0            0.0
```

```
# Anas platyrhynchos      0.05      0.60      0.35

# 4. Assign habitat to sampling sites based on species composition
occurrences_with_habitat <- assign_habitat_biological(
  occurrences, habitat_weights
)

# 5. Flag spatial outliers (marine species at inland sites, etc.)
flagged <- flag_habitat_inconsistencies(occurrences_with_habitat)
```

Key Functions

Habitat classification: - `build_habitat_prompt()` - create LLM prompt for species habitat weights - `parse_hierarchical_habitat_response()` - parse LLM output to numeric weights - `assign_habitat_biological()` - assign site habitat from species composition - `consensus_habitat()` - assemblage-level consensus with ecoregion extraction

Custom schemes: - `build_iucn_scheme()` - generate IUCN Level 1 habitat scheme - `example_habitat_scheme()` - example custom scheme for reference - `build_scheme_prompt()` / `parse_scheme_response()` - custom scheme workflow

Spatial QC: - `flag_habitat_inconsistencies()` - flag records inconsistent with site habitat - `review_spatial_flags()` - interactive Leaflet map for manual review

Reporting: - `report_habitat()` - summarize habitat assignment for `assemble_report()`

Methods

Habitat Weight Estimation

Rather than assign each species to a single habitat, TaxaHabitat asks the LLM to distribute habitat affinity as continuous weights across all habitat categories (e.g., Marine 0.85, Freshwater 0.15, Terrestrial 0.0), summing to 1.0 per species. This captures habitat generalism – an estuarine fish contributes partial signal to both Marine and Freshwater – and avoids the information loss of a categorical assignment.

Prompts are constructed by `build_habitat_prompt()` and sent to an LLM provider via TaxaTools. For large species lists, taxa are chunked (default 60 per call) into self-contained prompts. The LLM returns CSV with numeric weights per habitat plus an `Other_weight` column and free-text `habitat_best_guess` for species that do not fit the scheme. `parse_hierarchical_habitat_response()` validates that row sums are within tolerance (warns if deviation > 0.05 from 1.0) and folds unrecognized habitat columns into `Other_weight`.

Site Habitat Assignment

`assign_habitat_biological()` assigns a habitat to each sampling location based on the species observed there. For each point, the function joins occurrence records to species-level habitat weights, sums weight vectors across species, normalizes to proportions, and assigns the habitat with the highest proportion if it exceeds a threshold (default 0.3). The threshold is lower than a simple majority (0.5) because generalist species spread weight across multiple habitats, diluting any single category.

By default, each species counts equally regardless of how many times it was recorded at a point (`weight_by_abundance = FALSE`). This is deliberate: occurrence record counts reflect sampling effort, not ecological dominance.

Assemblage-Level Consensus

`consensus_habitat()` infers site habitat from a species list alone (without occurrence coordinates), using the same weighted-voting logic. When an `ecoregion_best_guess` column is present (from LLM output with `geographic_context`), the modal ecoregion across species is returned alongside the habitat consensus.

Spatial Quality Control

`flag_habitat_inconsistencies()` checks whether each occurrence record is spatially consistent with its assigned habitat using vector polygons (Natural Earth land/ocean boundaries) and raster bathymetry (GEBCO). Each record is classified into a physical zone:

Zone	Definition
inland	Inside land polygon, > 1 km from coast
coastal	Within 1 km of coastline (either side)
marine_shallow	Ocean, 0–200 m depth (continental shelf)
marine_deep	Ocean, 200–4000 m depth (bathyal)
marine_abyssal	Ocean, > 4000 m depth

Records are then cross-referenced against the habitat’s expected realm (marine, freshwater, terrestrial) and flagged at three levels: “likely” (consistent), “questionable” (borderline, e.g., marine species very near shore), or “unlikely” (clear mismatch, e.g., terrestrial species in open ocean). The 1 km coastal buffer accounts for GPS uncertainty and tidal gradients. `review_spatial_flags()` provides an interactive Leaflet map for manual inspection and correction.

LLM Integration

TaxaHabitat uses the `llm_fn` pattern from TaxaTools. The default provider is `call_anthropic_api()`, but any compatible provider works:

```
# Use Gemini instead
raw_text <- TaxaTools::prompt_api(prompt, llm_fn = TaxaTools::call_gemini_api)
```

API Keys

Requires an LLM API key (Anthropic by default). See the TaxaTools API Setup vignette for configuration.

Vignettes

- Habitat Assignment – full workflow guide

Part of TaxaID

TaxaHabitat receives occurrence data from TaxaFetch and produces habitat-annotated records for TaxaExpect (prior estimation).

Ecosystem: TaxaTools -> TaxaFetch -> **TaxaHabitat** -> TaxaExpect -> TaxaAssign

See the TaxaID README for ecosystem overview and installation instructions.

Citation

Lafferty, K.D., 2026, TaxaID – A modular R ecosystem for Bayesian taxonomic assignment: U.S. Geological Survey software release, <https://doi.org/10.5066/xxxxxx>.

Software Requirements

- R ($\geq 4.1.0$)
- TaxaTools (for LLM provider functions)
- An LLM API key is required for habitat assignment via `build_habitat_prompt()`

All dependencies are declared in the DESCRIPTION file and installed automatically.

Developed with Claude Code (Anthropic).